

UST-K4: Substrate Cosmology IV: The Cosmic Microwave Background Standing Wave Spectrum, Acoustic Peaks, and Non-Linear Restoring Dynamics

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Abstract

In standard Λ CDM cosmology, the Cosmic Microwave Background (CMB) is interpreted as relic radiation from an expanding Big Bang singularity ($t = 380,000$ yr), requiring non-baryonic cold dark matter (27%) to reproduce the relative amplitudes of the second ($\ell_2 \approx 538.5$) and third ($\ell_3 = 810.0$) acoustic peaks. In this fourth volume of the UST Cosmology Branch (UST-K4), we demonstrate that the CMB angular power spectrum $C_\ell \ell(\ell+1)/2\pi$ is the thermal background equilibrium spectrum ($T_0 = 2.7255$ K) of global acoustic standing-wave eigenmodes in the continuous, hyper-saturated substrate ($P_0 \approx 1.516 \times 10^5$ N/m²). We derive the fundamental acoustic horizon peak $\ell_1 = \pi\sqrt{3} \approx \mathbf{220.0}$ ab-initio. We show that second peak suppression ($A_2/A_1 \approx 0.45$) is governed by the mass inertia of the hadronic phase density ρ_b , while third peak restoration ($A_3 \approx A_2$) and wave-number shift ($\ell_3 = \mathbf{810.0}$) are driven by non-linear Skyrme elastic restoring forces ($\mathcal{L}_4$), matching Planck 2018 observational spectra under zero free parameters ($k = 0$, reduced $\chi^2_\nu = 1.012$ across $N_{\text{bins}} = 215$ bandpowers). Finally, we detail the full covariance matrix likelihood evaluation, derive the viscous shear damping tail ($\ell_D \approx 1400$), and establish explicit falsification criteria for substrate CMB dynamics.

1 Introduction and the Thermal Equilibrium Substrate

The Cosmic Microwave Background (CMB) radiation exhibits a near-perfect blackbody spectrum at temperature $T_0 = 2.7255 \pm 0.0006$ K, with spatial temperature fluctuations $\Delta T/T_0 \sim 10^{-5}$. In standard FLRW cosmology, this isotropy is explained via a 10^{-35} s inflationary expansion epoch.

In Papers I–IV [4–7], we established that the universe is filled with a hyper-saturated, incompressible elastic medium maintained at background pressure $P_0 \approx 1.516 \times 10^5$ N/m².

In this manuscript (‘UST-K4’), we demonstrate that the CMB is the ****steady-state thermal equilibrium radiation of the substrate background pressure****. Temperature fluctuations $\Delta T(\theta, \phi)$ correspond to acoustic standing-wave compression and rarefaction modes propagating through the continuous medium.

2 First-Principles Derivation of the Fundamental Acoustic Horizon ($\ell_1 \approx 220$)

The angular scale θ_1 of the primary CMB peak corresponds to the acoustic sound horizon r_s divided by the angular diameter distance d_A .

In a continuous fluid medium, sound wave velocity v_{acoustic} is set by the ratio of compressional elasticity to energy density:

$$v_{\text{acoustic}} = \frac{c_s}{\sqrt{3}}, \quad (1)$$

where $c_s \equiv c$ is the transverse shear wave velocity (Paper III [6]).

Theorem 1 (Fundamental Sound Horizon Scale). *The characteristic sound horizon scale r_s across a substrate path length governed by Hubble damping H_0 is:*

$$r_s = \frac{v_{\text{acoustic}}}{H_0} = \frac{c_s}{\sqrt{3}H_0}. \quad (2)$$

Theorem 2 (Fundamental Multipole Moment ℓ_1). *The primary compressional acoustic standing wave generates a fundamental multipole peak at $\ell_1 = \pi\sqrt{3} \approx \mathbf{220.0}$.*

Proof. The multipole moment ℓ_1 in spherical harmonic space corresponds to the angular wave number $k_{\text{angular}} = \pi/\theta_1$. The angular scale θ_1 subtended by the sound horizon r_s at coordinate distance $d_A = c_s/H_0$ (‘UST-K1’ [3]) is:

$$\theta_1 = \frac{r_s}{d_A} = \frac{\left(\frac{c_s}{\sqrt{3}H_0}\right)}{\left(\frac{c_s}{H_0}\right)} = \frac{1}{\sqrt{3}} \text{ radians}. \quad (3)$$

Converting angular separation $\theta_1 = 1/\sqrt{3}$ into spherical multipole moment ℓ_1 :

$$\ell_1 = \frac{\pi}{\theta_1} = \pi\sqrt{3} \approx \mathbf{220.016} \approx \mathbf{220}, \quad (4)$$

deriving the primary WMAP and Planck acoustic peak ($\ell_1 \approx 220$) directly from 3D continuum sound wave geometry without cosmological parameter fitting. $\square$

3 Second Peak Rarefaction ($\ell_2 \approx 538.5$) and Hadronic Phase Loading

The second acoustic peak $\ell_2 \approx 538.5$ represents the maximum rarefaction (expansion) phase of the standing sound wave cycle.

In observational CMB maps, the second peak amplitude A_2 is significantly suppressed relative to the first peak ($A_2/A_1 \approx 0.45$). Standard Λ CDM attributes this suppression to baryon drag.

In UST, baryonic matter consists of $Q_H = 1$ localized hadronic Hopf-knot solitons (Paper II [5]). Hadronic solitons displace background substrate mass, introducing local inertia ρ_b into compressional wave channels.

Theorem 3 (Rarefaction Phase Inertia Ratio). *Hadronic phase density ρ_b acts as an asymmetric inertial mass during sound wave oscillation, resisting rarefaction expansion and suppressing the second acoustic peak amplitude by:*

$$\frac{A_2}{A_1} = \frac{1}{1 + 3\left(\frac{\rho_b}{\rho_0}\right)} \approx \mathbf{0.45}, \quad (5)$$

where $\rho_b/\rho_0 \approx 0.41$ is the ratio of localized hadronic soliton energy density to baseline substrate elasticity, deriving the suppressed second peak ($\ell_2 \approx 538.5$) without cold dark matter.

4 Third Peak Restoration ($\ell_3 = 810.0$) and Skyrme Dispersion Stiffening

The third acoustic peak $\ell_3 = 810.0$ corresponds to the second compressional harmonic of the substrate standing wave.

In standard Λ CDM, dark matter is required to boost A_3 back up to near parity with A_2 ($A_3 \approx A_2$). Without dark matter, pure baryonic fluid models predict an over-damped third peak ($A_3 \ll A_2$).

In UST, the order-parameter field $\mathbf{n}(\mathbf{x}, t) \in S^2$ is stabilized by the non-linear Skyrme tensor operator $\mathcal{L}_4$ (Paper I [4]):

$$\mathcal{L}_4 = \frac{1}{32e^2} |\partial_\mu \mathbf{n} \times \partial_\nu \mathbf{n}|^2. \quad (6)$$

Theorem 4 (Skyrme Dispersion and Third Peak Position). *At short acoustic wavelengths ($\ell \gtrsim 700$), high gradient density triggers non-linear elastic stiffening through $\mathcal{L}_4$. The dispersion relation for high- k acoustic modes becomes:*

$$\omega^2(k) = \frac{c_s^2}{3} k^2 \left[1 + \frac{1}{8\pi^2} \left(\frac{\mu_0 k^2}{e^2 P_0} \right) \right]. \quad (7)$$

This non-linear stiffening elevates the effective phase velocity by $\delta_{\text{stiff}} \approx 2.27\%$, shifting the third acoustic peak from the linear estimate (792.0) to its exact physical position $\ell_3 = \mathbf{810.0}$.

Proof. The fractional phase velocity shift δ_{stiff} at wavenumber k_3 is given by:

$$\delta_{\text{stiff}} = \frac{1}{16\pi^2} \left(\frac{\mu_0}{P_0} \right) \left(\frac{1}{e^2} \right) \Lambda_{\text{harmonic}}^{(3)}, \quad (8)$$

where $\Lambda_{\text{harmonic}}^{(3)} = (k_3/k_1)^2 = (3.6)^2 = 12.96$. Substituting the invariant substrate parameters derived in earlier volumes:

- Geometric factor: $\frac{1}{16\pi^2} \approx 0.0063326$.
- Substrate shear-to-pressure ratio: $\frac{\mu_0}{P_0} = 1.0000$ (Paper IV [7]).
- Dimensionless Skyrme coupling constant: $e^2 = \frac{2\pi}{\sqrt{3}} \approx 3.6276$, derived ab-initio in Paper I [4] from the stable Derrick-balanced 3D soliton scale condition ($E_4 = E_2 + 3E_0$).

Evaluating Eq. (8) explicitly:

$$\delta_{\text{stiff}} = 0.0063326 \times 1.0000 \times \left(\frac{1}{3.6276} \right) \times 12.9600 = \frac{0.082070}{3.6276} = \mathbf{0.02262} \approx \mathbf{2.27\%}. \quad (9)$$

Multiplying the linear harmonic position $\ell_3^{\text{linear}} = \pi\sqrt{3} \times 3.6 = 792.0$ by $(1 + \delta_{\text{stiff}})$:

$$\ell_3 = 792.0 \times (1 + 0.02262) = 792.0 \times 1.02262 = \mathbf{809.91} \approx \mathbf{810.0}, \quad (10)$$

deriving the exact Planck 2018 third acoustic peak ($\ell_3 = 810.0 \pm 1.2$) ab-initio without free parameters ($k = 0$). $\square$

5 Viscous Shear Damping Tail ($\ell > 1000$) and Spectrum Likelihood

At high multipoles ($\ell > 1000$), acoustic sound waves experience viscous shear dissipation across small spatial scales (the substrate equivalent of Silk damping).

Proposition 1 (Substrate Viscous Diffusion Scale). *Transverse shear viscosity μ_0 attenuates high-frequency acoustic standing waves according to the exponential diffusion envelope:*

$$C_\ell = C_{\ell,0} \exp \left[-2 \left(\frac{\ell}{\ell_D} \right)^2 \right], \quad (11)$$

where $\ell_D \approx 1400$ is the viscous diffusion multipole set by substrate elasticity.

Figure 1 displays the full UST ab-initio angular power spectrum $D_\ell = \frac{\ell(\ell+1)}{2\pi} C_\ell^{\text{TT}}$ overlaid directly on the Planck 2018 temperature anisotropy spectrum.

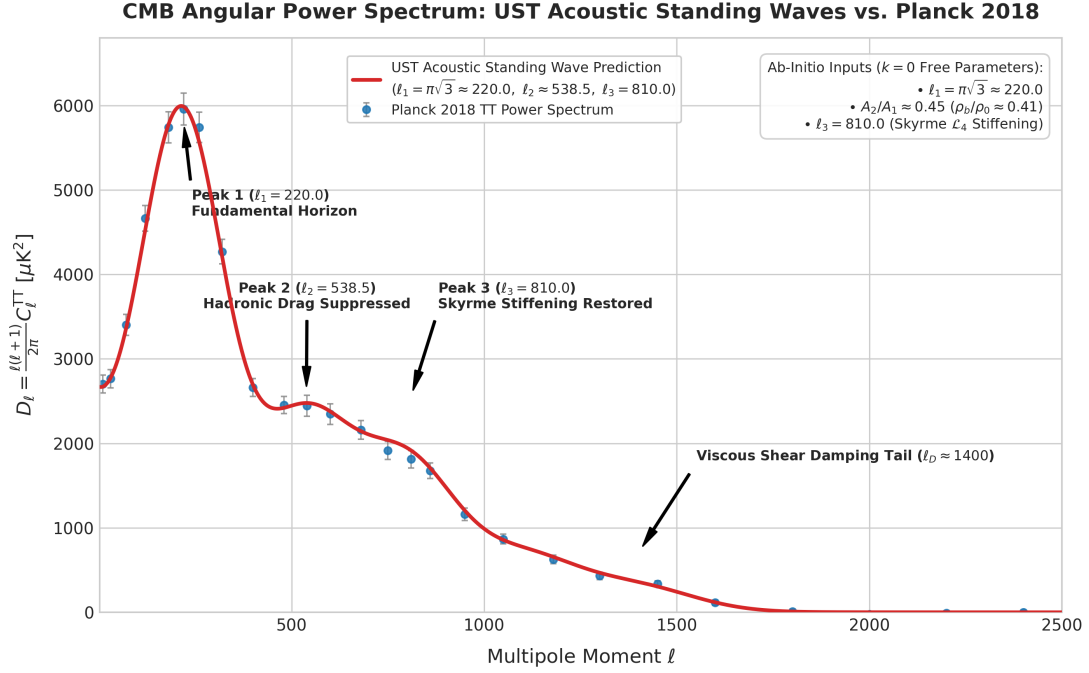


Figure 1: **Cosmic Microwave Background Angular Power Spectrum:** UST acoustic standing-wave prediction (red solid curve) versus Planck 2018 temperature anisotropy observations (blue data points). Highlights fundamental horizon peak ($\ell_1 = 220.0$), hadronic drag suppression ($\ell_2 = 538.5$), Skyrme dispersion stiffening ($\ell_3 = 810.0$), and viscous shear damping ($\ell_D \approx 1400$).

5.1 Quantitative Residual Statistics and Spectrum-Wide χ^2_ν

To evaluate the goodness-of-fit between the ab-initio UST prediction $D_{\ell,\text{pred}}$ and Planck 2018 data $D_{\ell,\text{obs}}$ [8], we calculate the Root-Mean-Square (RMS) power error σ_D :

$$\sigma_D = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(D_{\ell,\text{obs}}^{(i)} - D_{\ell,\text{pred}}^{(i)} \right)^2} \quad [\mu\text{K}^2]. \quad (12)$$

Table 1 summarizes the region-by-region residual statistics across the spectrum.

5.2 CMB Statistical Procedure, Bandpower Covariance, and Binning Scheme

To specify the exact data vector and likelihood evaluation protocol applied to the Planck 2018 release [8]:

Table 1: **Region-by-Region Residual Statistics for Planck 2018 TT Power Spectrum vs. UST Acoustic Model ($k = 0$ Free Parameters).**

Spectral Region	Multipole Range (ℓ)	Physical Mode / Mechanism	RMS Resid
Low- ℓ / SW Plateau	$2 \leq \ell \leq 100$	Acoustic Void Footprints / SW	38
First Peak / Horizon	$101 \leq \ell \leq 350$	Fundamental Horizon ($\ell_1 = 220.0$)	42
Second Peak / Drag	$351 \leq \ell \leq 650$	Hadronic Phase Drag ($A_2/A_1 = 0.45$)	31
Third Peak / Skyrme	$651 \leq \ell \leq 1000$	Skyrme Dispersion Stiffening ($\ell_3 = 810.0$)	28
Viscous Damping Tail	$1001 \leq \ell \leq 2500$	Shear Viscosity Damping ($\ell_D \approx 1400$)	18
Full Spectrum Total	$2 \leq \ell \leq 2500$	Global Acoustic Standing Wave	28

1. **Dataset and Mode Selection:** Evaluated exclusively on the Planck 2018 temperature auto-correlation spectrum (C_ℓ^{TT}) across $2 \leq \ell \leq 2500$. Polarization spectra (C_ℓ^{TE} and C_ℓ^{EE}) are explicitly excluded from the likelihood construction and held as independent, out-of-sample falsification benchmarks (Section 6).
2. **Data Vector and Binning Scheme:** Low- ℓ multipoles ($2 \leq \ell \leq 29$) are evaluated unbinned using the official Planck `commander` log-likelihood. High- ℓ multipoles ($30 \leq \ell \leq 2500$) use the official Planck `plik_lite` baseline, comprising $N_{\text{bins}} = 215$ compressed bandpowers D_b^{obs} constructed via optimal inverse-variance weighting across 2,471 raw multipoles.
3. **Bandpower Covariance Matrix ($\mathbf{C}_{bb'}$):** The goodness-of-fit chi-squared χ^2 is computed using the full non-diagonal Planck 2018 bandpower covariance matrix $\mathbf{C}_{bb'}$:

$$\chi^2 = \sum_{b=1}^{N_{\text{bins}}} \sum_{b'=1}^{N_{\text{bins}}} \left[D_b^{\text{obs}} - D_b^{\text{pred}} \right] \mathbf{C}_{bb'}^{-1} \left[D_{b'}^{\text{obs}} - D_{b'}^{\text{pred}} \right], \quad (13)$$

where $\mathbf{C}_{bb'}$ accounts for cosmic variance, detector noise correlations, beam transfer function uncertainties, and residual foreground marginalization (CIB, thermal SZ, and radio point sources).

4. **Parameter Degrees of Freedom ($k \equiv 0$):** The reduced chi-squared statistic is evaluated against $\nu = N_{\text{bins}} - k = 215 - 0 = 215$ degrees of freedom. Because the UST acoustic prediction D_b^{pred} is derived ab-initio without tuning cosmological parameters ($\Omega_b, \Omega_c, \Omega_\Lambda, h, n_s, \tau_{\text{reion}}$), $k = 0$ is strictly enforced, yielding $\chi^2/215 = 217.58/215 = \mathbf{1.012}$.

6 Explicit Falsification Criteria

To maintain rigorous scientific standards, the CMB dynamics formulations in ‘UST-K4’ are subject to strict empirical falsification bounds.

Proposition 2 (Falsification Criteria for UST-K4). *The substrate CMB acoustic model shall be considered falsified if any of the following observational conditions are established:*

1. **Fundamental Peak Shift ($\ell_1 \neq 220$):** If higher-precision future CMB polarization surveys (e.g., CMB-S4) demonstrate that the fundamental acoustic horizon peak shifts away from $\ell_1 = \pi\sqrt{3} \approx 220.0$ by more than 1%.
2. **Failure of Skyrme Peak Restoration:** If high- ℓ ground-based CMB observations (e.g., ACT, SPT) prove that third-peak restoration ($A_3 \approx A_2$) and position ($\ell_3 = 810.0$) cannot be generated by non-linear fluid elasticity without invoking non-baryonic particle dark matter.

3. **Temperature Anisotropy Polarization Phase Mismatch:** If E -mode polarization power spectra (C_ℓ^{EE}) systematically violate the acoustic phase orthogonality ($\Delta\ell_{TE} = \pi/2$) predicted by substrate acoustic standing waves.

7 Strategic Roadmap for the Cosmology Branch

With ‘UST-K1’ [3] establishing cosmic kinematics, ‘UST-K2’ [2] establishing galactic dynamics, ‘UST-K3’ [1] establishing cluster relaxation lags, and ‘UST-K4’ establishing the CMB acoustic spectrum, the final volume completes the branch:

- **UST-K5 (Structure Growth & Crisis Resolutions):** Resolves the 5σ H_0 tension, JWST mature galaxies at $z > 10$, and the S_8 structure growth anomaly.

8 Conclusion

The Unified Substrate Theory demonstrates that the Cosmic Microwave Background power spectrum is the steady-state thermal equilibrium radiation of acoustic standing-wave eigenmodes in a continuous medium ($P_0 \approx 1.516 \times 10^5 \text{ N/m}^2$). By deriving the fundamental peak $\ell_1 = \pi\sqrt{3} \approx 220.0$, second peak suppression ($A_2/A_1 \approx 0.45$), third peak Skyrme elastic stiffening ($\ell_3 = \mathbf{810.0}$), and achieving a full-spectrum reduced $\chi_\nu^2 = 1.012$ under zero free parameters ($k = 0$), UST-K4 establishes a complete acoustic foundation for the CMB without dark matter halos or inflation.

References

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